

MATTHEW H. TAYLOR

🔗 mht3.github.io

Research Focus: AI for scientific discovery, physical applications of deep learning and reinforcement learning.

EDUCATION

UNIVERSITY OF CALIFORNIA, SAN DIEGO

- PhD in Computer Science and Engineering (Est. Graduation: 06/29)
- MS in Computer Science and Engineering

GPA: 4.0 / 4.0

SEP. 2025 - PRESENT

SEP. 2023 - JUNE 2025

UNIVERSITY OF ILLINOIS AT URBANA-CHAMPAIGN

- Bachelor of Science in Aerospace Engineering
- Highest Honors, Minor in Computer Science

GPA: 3.93 / 4.00

AUG. 2019 - MAY 2023

WORK EXPERIENCE

NASA Jet Propulsion Laboratory, Machine Learning Research Intern JUNE 2024 - NOV. 2024

- Leveraged self-supervised contrastive learning techniques to create an ocean synthetic-aperture radar (SAR) foundation model and finetuned on ocean science object detection and semantic segmentation tasks.
- Developed a versatile, dataset and model-agnostic semantic segmentation and object detection training codebase with integration of Weights and Biases for visualizing and monitoring model performance.
- Curated a dataset with bounding box and polygon mask annotations for over 7,000 submesoscale ocean eddies in the western Mediterranean Sea and prepared for NASA PO.DAAC (NASA oceanography data archive).

Dolby Laboratories, Computer Vision/ML Engineer Intern JUNE 2023 - AUG. 2023

- Profiled video content using skin segmentation, noise, face, and text detection models to give suggestions for when to use precision detail, a local image adjustment algorithm created for Dolby Vision.
- Updated Dolby Vision's internal C++ to Python bridge by adding hooks for local enhancement parameters and produced a modular PyQt GUI to visualize content mapped Dolby Vision content and recommendations.

Maxar Technologies, Research and Development Intern MAY 2022 - AUG. 2022

- Applied deep learning and semantic segmentation methods to detect building footprints from Maxar satellite imagery using PyTorch Lightning for training and TensorBoard for model evaluation and visualizations.
- Researched how the addition of the near infra-red band (NIR) improves segmentation performance.

RESEARCH AND TEACHING EXPERIENCE

Graduate Researcher, Advised by Professor Sicun Gao MARCH 2024 - PRESENT

- Improving learning-based design for high dimensional control problems using ideas from Lyapunov stability, imitation learning, and learning from imperfect experts.
- Creating more stable, robust policies for the Unitree G1 humanoid robot for balance and locomotion tasks.
- Created an approximate neural Lyapunov controller for pendulum and cartpole gymnasium environments with the goal of expanding and generalizing to problems with larger state spaces and unknown dynamics.

Teaching Assistant, UC San Diego

- CSE 8A: Intro to Programming JAN. 2024 - MARCH 2024
- CSE 257: Search and Optimization JAN. 2025 - MARCH 2025
- CSE 150B: AI Search and Reasoning APRIL 2025 - JUNE 2025

Undergraduate Researcher, Tran Research Group MAY 2021 - MAY 2023

- Created a CNN to detect objects and transform a robot's camera view into a top-down grid world in simulation and the real-world using the NVIDIA JetBot AI kit. Controlled the robot in simulation using Gazebo and ROS.
- Explored useful applications for brain computer interfaces and human-agent teaming, creating an experiment to classify LEDs flashing at different frequencies based on EEG signals to improve robot collaboration.

SKILLS

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|----------|---------|------------|--------------|
| • Python | • C/C++ | • Java | • PyTorch |
| • Git | • Unix | • IsaacLab | • TensorFlow |

RELEVANT COUSEWORK

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|---------------------------|------------------------------|----------------------------|
| • Search and Optimization | • Random Processes | • Introduction to AI |
| • Algorithms | • Probability and Statistics | • Introduction to Robotics |
| • ML for Physical Appl. | • Parallel Computing | • Machine Learnin |